

Exercise: Decoding (100 Points)

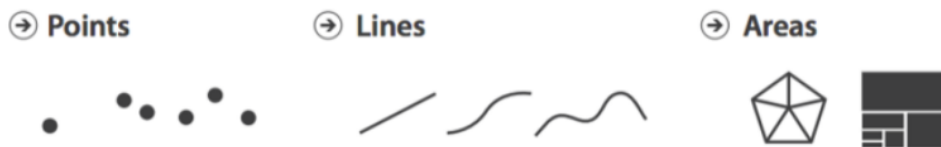
Learning Goal: In this exercise you will practice breaking down charts into their most basic elements: marks and channels. Virtually every single visualization can be described using this language, according to the rules of *visual encoding*.

Review of main concepts

In class (W4) we discussed the idea that every visualization can be described as

- 1) A mapping between **data items** (e.g. a row in your dataset) and **visual marks (points, lines, areas)** and
- 2) A mapping between **data attributes** (e.g. a column or variable in your dataset) and **visual channels (see below)**.

Marks



We distinguish between two different types of channels. The first are **identity channels**, which we use to represent categorical variables (e.g. names, location information like country or state, species of a flower, etc.). These variables do not have an inherent order. For instance, you could order countries alphabetically or by GDP - neither is inherently more correct, and the choice depends on your data questions. We use identity channels like spatial region, color hue, motion, and shape to represent these variables.

Identity Channels

Tell us information about **what** something is or **where** it is

➔ Identity Channels: Categorical Attributes

Spatial region



Color hue



Motion

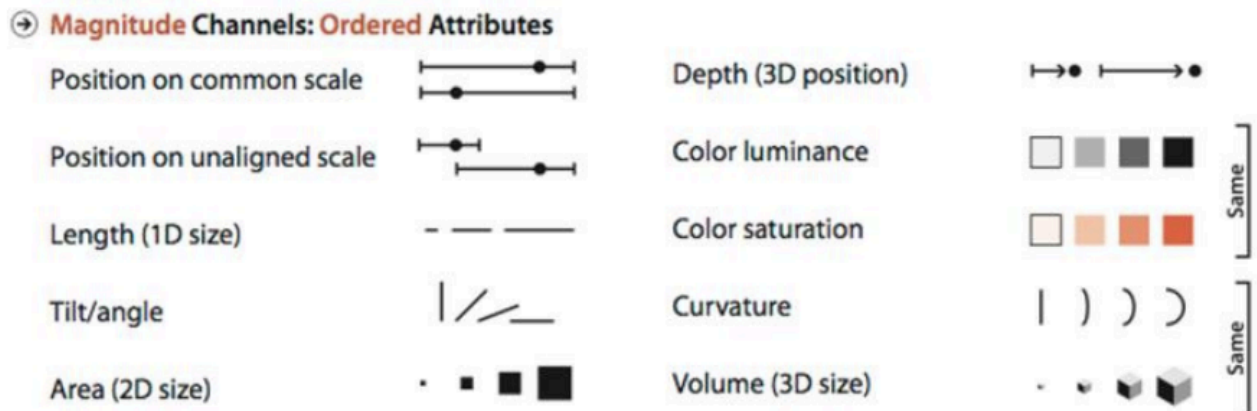


Shape



Magnitude Channels

Tell us **how much** of something there is



The second type of channels are **magnitude channels**. We use these to represent everything else - quantitative, ordered variables.

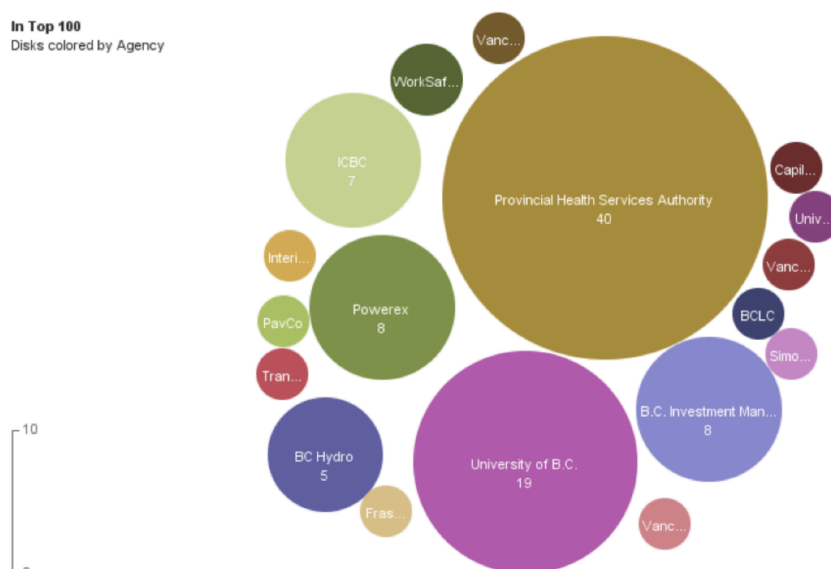
Remember that color is not a single channel. You need to specify which component of color encodes information from your data - color hue, color luminance, or color saturation.

Keep in mind that some visualizations may use one of these channels without encoding any of the attributes in the data with it. For instance, in a bubble chart like the one below, each bubble has a position, but position does not encode any specific information from the data!

Where B.C.'s 100 highest paid public servants work

Created at: May 23 2012

In Top 100
Disks colored by Agency

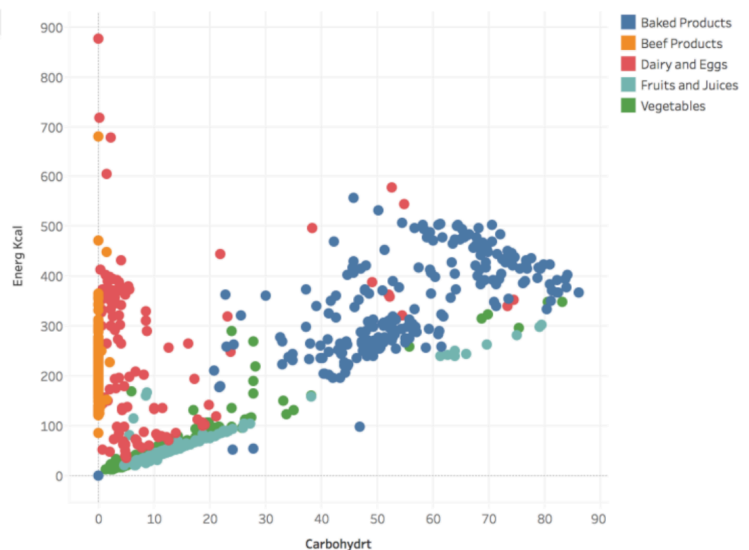


When this happens, you do not need to specify a visual encoding rule. In general, in any given visualization there are many graphical objects other than those that encode data directly (e.g. grid lines, tick marks, labels). In this exercise you only have to specify rules for visual marks and channels that directly encode information from the data.

Below is an example graphical decoding task with a scatter plot. The plot encodes the data depicted in the table on the left, wherein each row represents one food product.

Each data object is a *food product* and each food product is represented by a symbol mark (a dot). The graph uses three channels: x-position, y-position, and color hue. The x-position of each dot encodes the amount of carbohydrates, the y-position encodes the amount of calories (Energy Kcal), and the color hue encodes the food type (FdGrp Desc).

Carbohydrdt	Energ Kcal	FdGrp Desc
83.2800	349	Vegetables and Vegetable Products
80.7000	348	Vegetables and Vegetable Products
75.3700	296	Vegetables and Vegetable Products
69.8600	324	Vegetables and Vegetable Products
68.7000	314	Vegetables and Vegetable Products
55.7600	258	Vegetables and Vegetable Products
38.0600	160	Vegetables and Vegetable Products
35.0600	130	Vegetables and Vegetable Products
33.7300	122	Vegetables and Vegetable Products
33.0600	149	Vegetables and Vegetable Products
28.1000	218	Vegetables and Vegetable Products
27.8100	268	Vegetables and Vegetable Products
27.7500	190	Vegetables and Vegetable Products
27.7400	164	Vegetables and Vegetable Products
27.4800	116	Vegetables and Vegetable Products
26.9900	114	Vegetables and Vegetable Products
26.2300	112	Vegetables and Vegetable Products
25.7200	96	Vegetables and Vegetable Products
23.9400	97	Vegetables and Vegetable Products
23.9000	290	Vegetables and Vegetable Products
23.8800	136	Vegetables and Vegetable Products
21.7100	97	Vegetables and Vegetable Products
21.5500	93	Vegetables and Vegetable Products
21.4400	97	Vegetables and Vegetable Products
21.1500	93	Vegetables and Vegetable Products
21.1500	91	Vegetables and Vegetable Products



Instructions

For each of the following charts, identify:

- 1) The **visual marks** and the **data objects** they represent and
- 2) The **visual channels** they use and the **data attributes** they encode.

Use this template for each chart:

Data item: ...

Mark: ...

Magnitude Channels:

- <Channel type 1>: <Attribute name>
- ...
- <Channel type N>: <Attribute name>

Identity Channels:

- <Channel type 1>: <Attribute name>
- ...
- <Channel type N>: <Attribute name>

Using this template, the solution for the scatter plot discussed above would be

***Data Item:** food product*

***Mark:** point*

Magnitude Channels:

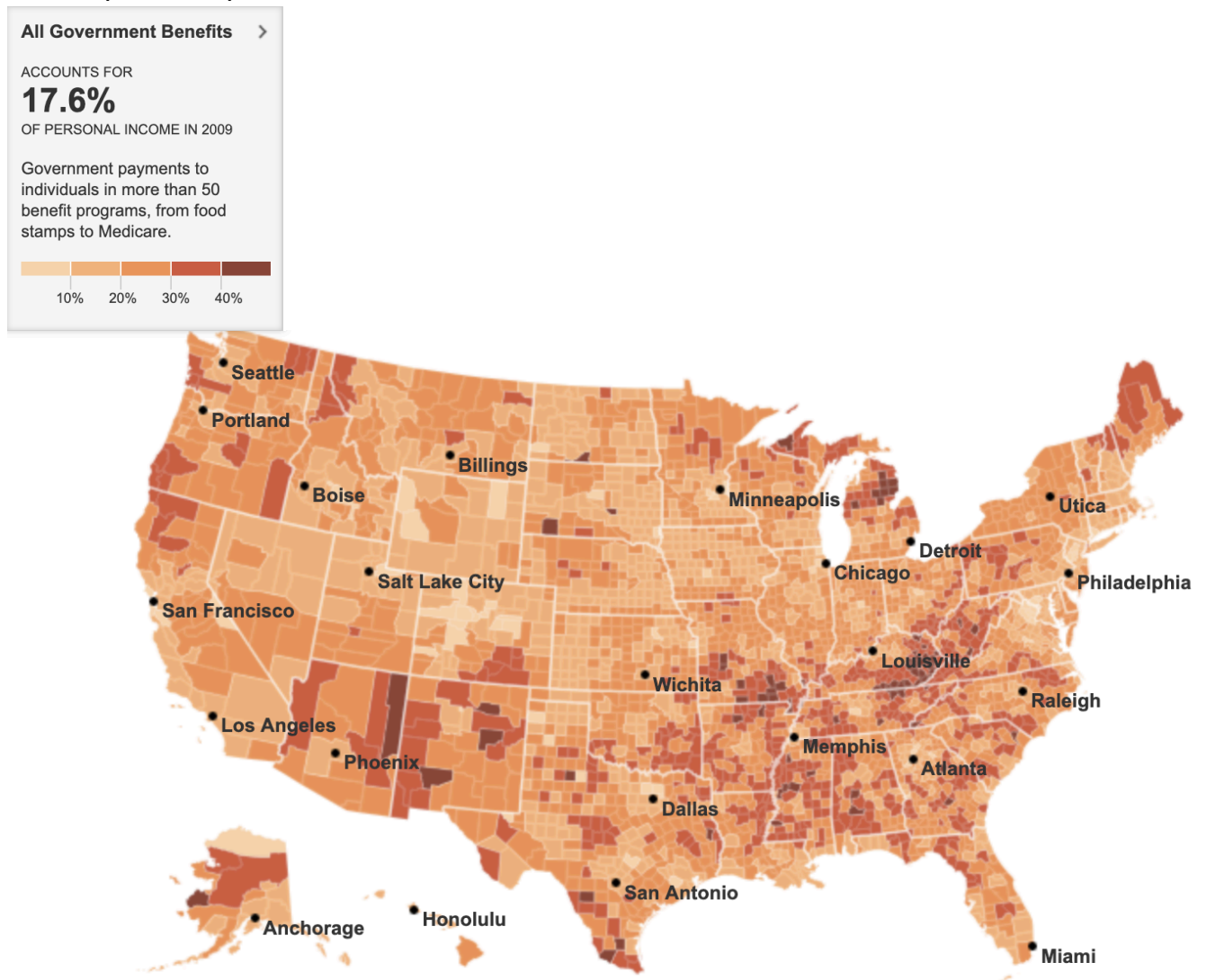
- *X-position: carbs*
- *Y-position: calories*

Identity Channels:

- *Color hue: food category*

NOTE: at the bottom of all charts there is a link to a web page providing details about the project (often including information about the data).

Chart 1 (20 Points)



<http://www.nytimes.com/interactive/2012/02/12/us/entitlement-map.html>

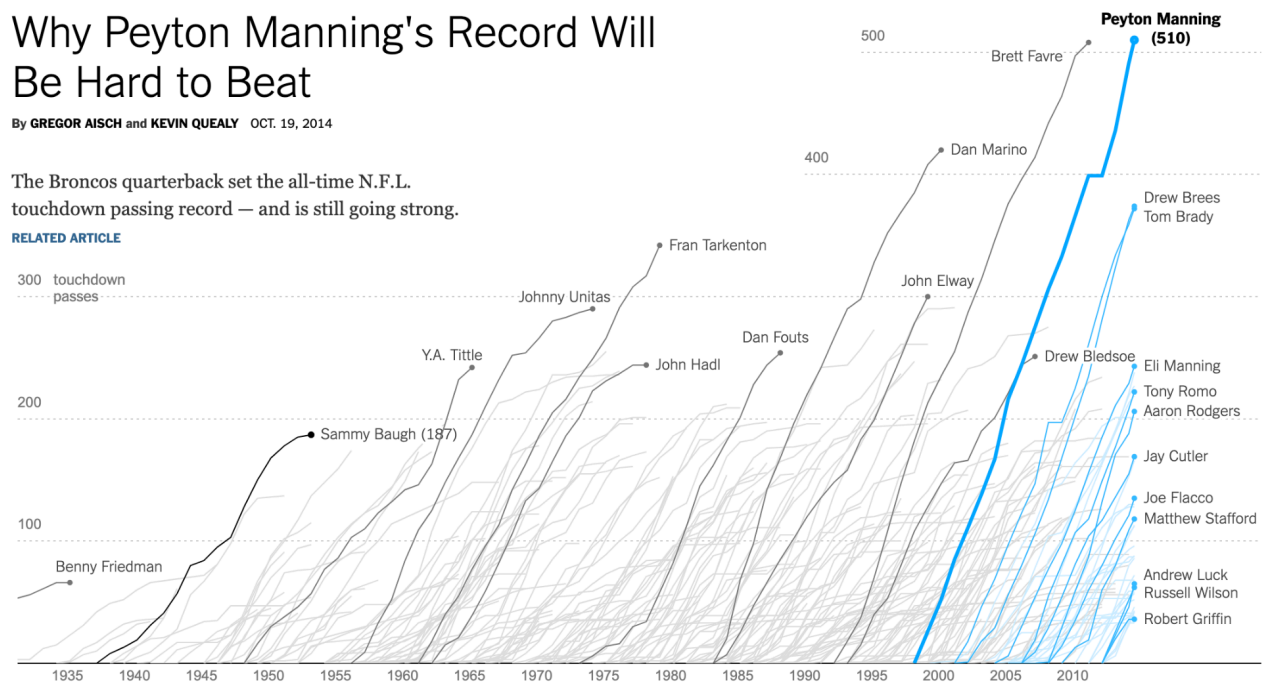
Chart 2 (20 Points)

Why Peyton Manning's Record Will Be Hard to Beat

By GREGOR AISCH and KEVIN QUEALY OCT. 19, 2014

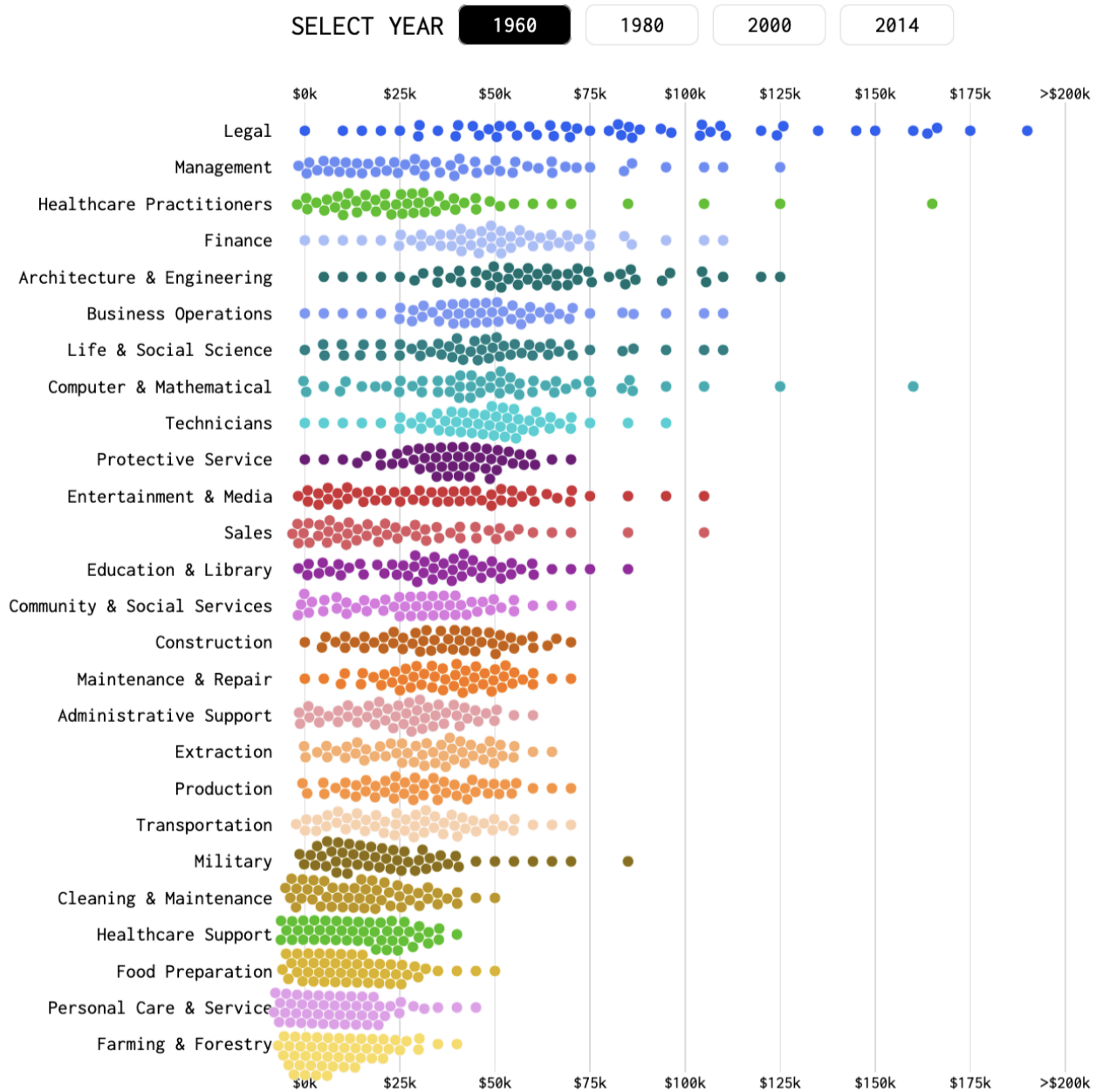
The Broncos quarterback set the all-time N.F.L. touchdown passing record — and is still going strong.

[RELATED ARTICLE](#)



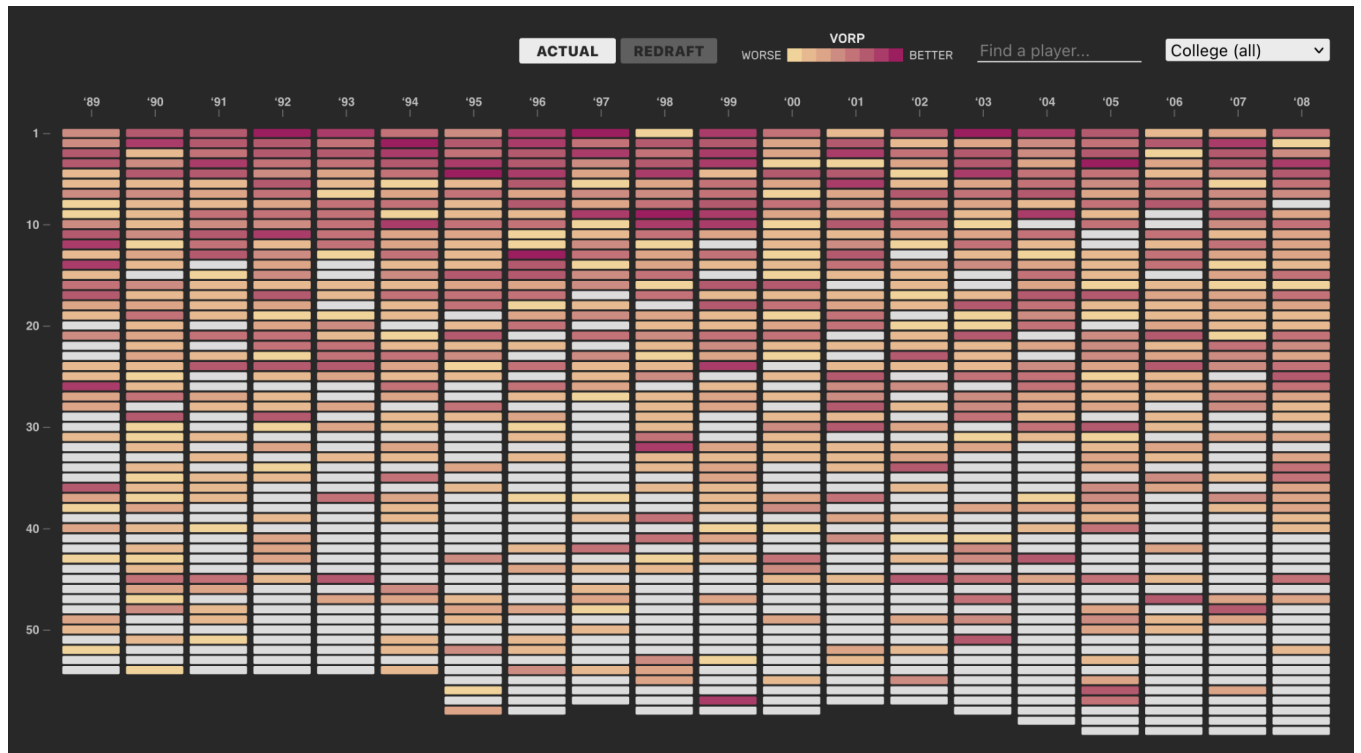
<https://www.nytimes.com/interactive/2014/10/19/upshot/peyton-manning-breaks-touchdown-passing-record.html>

Chart 3 (20 Points)



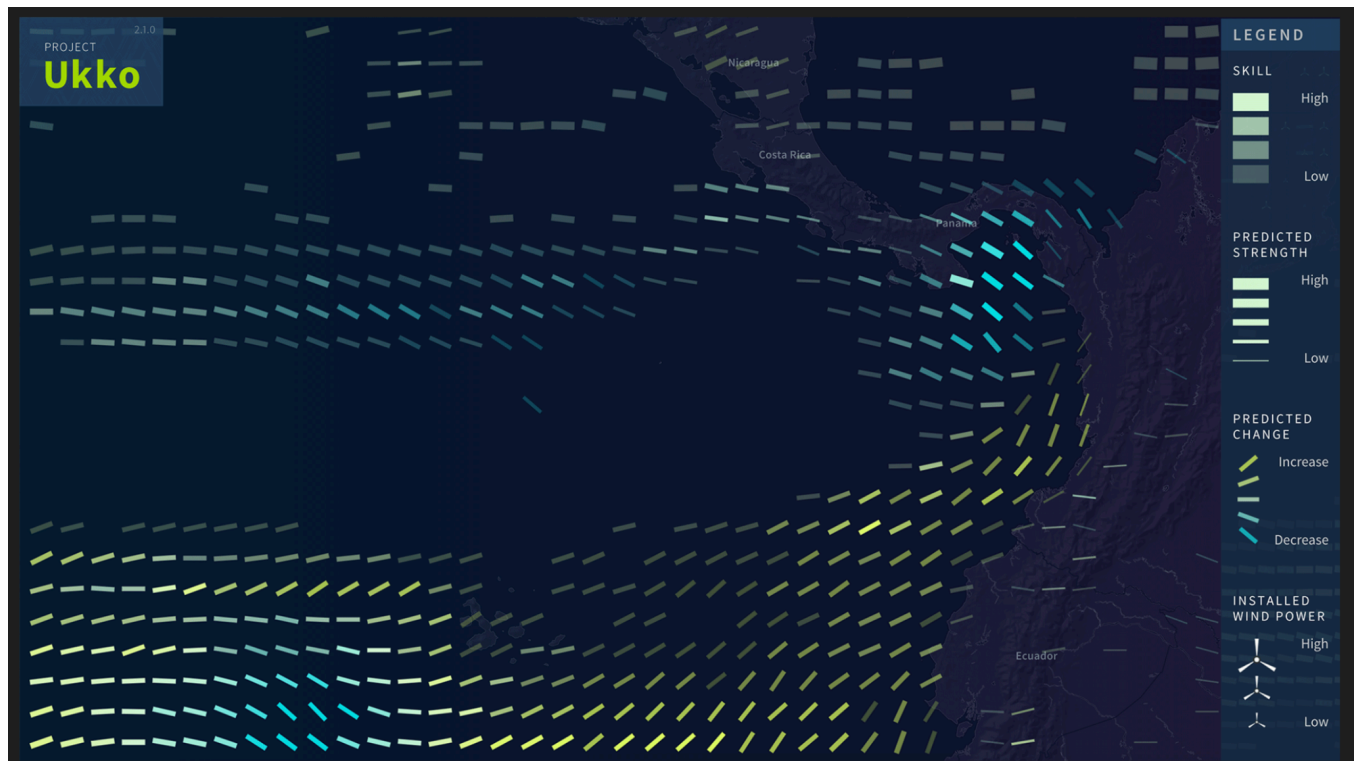
<https://flowingdata.com/2016/06/28/distributions-of-annual-income/>

Chart 4 (20 Points)



<https://pudding.cool/2017/03/redraft/>

Chart 5 (20 Points)



<http://project-ukko.net/> and <https://truth-and-beauty.net/projects/ukko>